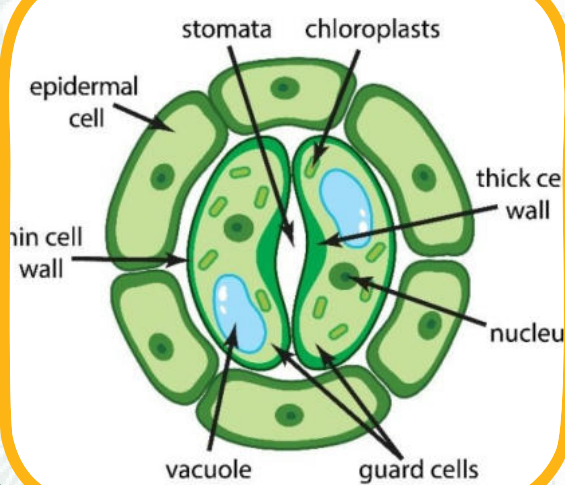




TRANSPORTATION



Transportation in Plants:

Roots:

The root of that part, which grows under the soil and develops the radicle of the seed. They absorb water & salts from soil, they provide conducting tissues for distributing these substances to the tissues of the stem.

Internal Structure:

Vascular Tissues:

The conducting tissues (xylem & phloem) of the root are grouped in the center to form a rod shaped cor. This rod extends throughout the root.

Pericycle:

Outside the conducting tissues, there is a narrow layer of thin-walled cells, the pericycle.

Endodermis:

A single layer of cells called endodermis surrounds the pericycle.

Cortex:

Eternal to endodermis, there is a broad zone of cortex. It consists of large and thin-walled cells.

Epidermis:

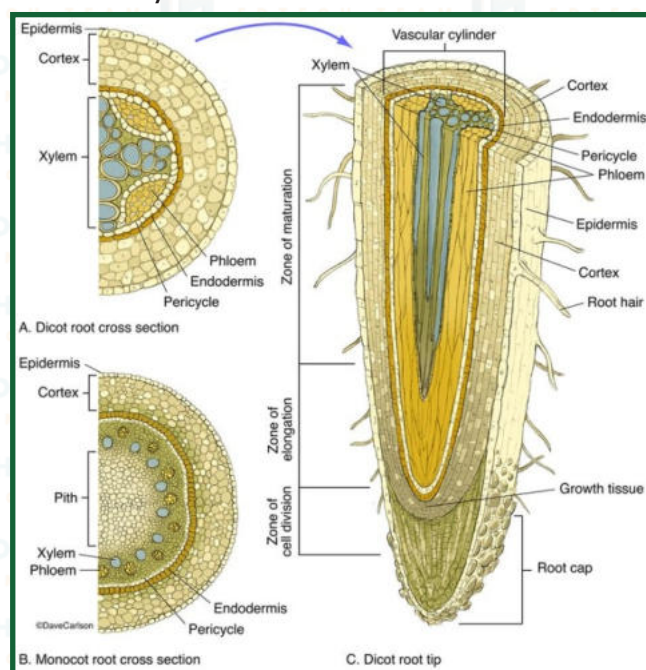
The cortex is bounded on the outside by a single layer of epidermal cells.

Root Hairs:

Roots have clusters of tiny root hairs, the extension of epidermal cells.

Function:

Root hairs provide a large surface area for reabsorption. They grow out into the spaces between soil particles where they are in direct contact with water.



Modes of Transport

There are two chief means of transportation of molecules across the cell membrane which are listed as under:

- i. Passive transport
- ii. Active transport

Passive Transport:

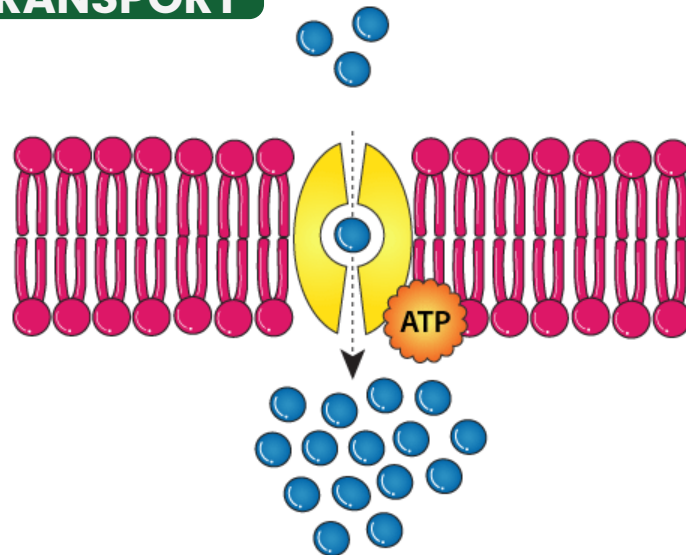
The uptake of water and minerals due to concentration gradient without using the energy of ATP is called passive transport. Passive transport always takes place from high to low quantities of substances.

Examples:

- Osmosis
- Diffusion

Active Transport:

Movement of substances from low quantity to high quantity i.e. against the concentration gradient which requires energy of ATP is called active transport.

ACTIVE TRANSPORT**Mineral Transportation in Plants**

Plant also requires minerals i.e. nitrates, sulphates, phosphate etc. These minerals are also taken up by root hair in two-ways:

- Diffusion
- Active transport

Diffusion:

It is an overall movement of molecules of a substance from high concentration towards low concentration till a uniform concentration (equilibrium) is achieved.

Transport of Minerals by Diffusion:

Minerals and certain ions transport from the soil into the root hair by diffusion, when the concentration of certain ions in soil is higher than that in root hair cells i.e. passive transport.

Active Transport:

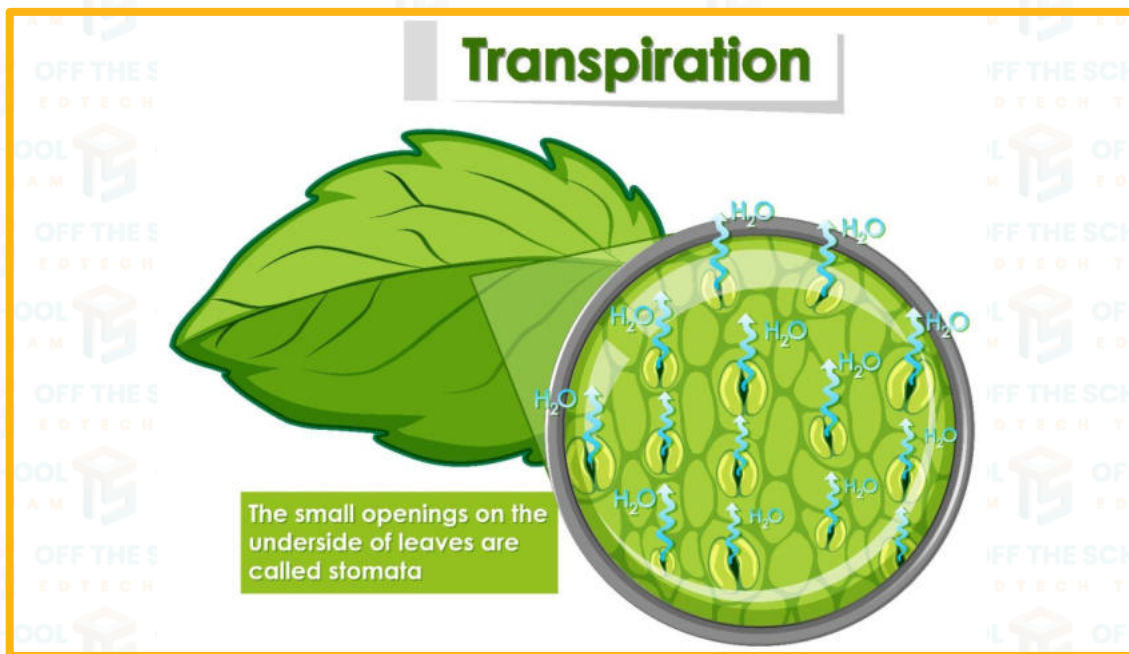
Movement of substances from low quantity to high quantity i.e. against the concentration gradient which requires energy of ATP is called active transport.

Transport of Minerals by Active Transport:

Plants require some substances even if they are found in soil in low quantities. The roots absorb these ions against a concentration gradient by using energy of ATP, which is active transport.

Transpiration**Definition:**

The evaporation of water from aerial parts of plants is known as transpiration.



Factors Affecting the Rate of Transpiration:

The rate of transpiration is affected by the following environmental factors.

Light:

It is one of the most important factors because it regulates the opening & closing of stomata. The stomata remain wide open during day time, as a result transpiration takes place through them but at night they are closed.

Temperature:

The increase in temperature increases the rate of transpiration by increasing the rate of evaporation of water from the cell surface.

Humidity:

Transpiration decreases with the increase in humidity (moisture of air) and increases with the decrease in humidity. It means that transpiration can only take place when the atmosphere is partially saturated or dry. The increase in the wind velocity increases the rate of transpiration through a reduction in the density of the air. Low atmospheric pressure increases the rate of transpiration through reduction in the density of the air.

Significance of Transpiration:

- 1) By active transpiration a suction forces the transpiration pull is created which helps in upward movement of water and minerals.
- 2) It also increases the rate of transpiration.
- 3) It helps the evaporation of excess amounts of water.
- 4) It maintains a suitable temperature and prevents overheating.
- 5) It facilitates the ascent of sap.
- 6) It brings about the closing and opening of stomata which indirectly influences the process of photosynthesis and respiration.
- 7) Excessive loss of water from aerial parts also results in dehydration leading to the health of plants in extreme condition.

Stomata: (Singular; Stoma)

Stomata are the special pore-like structure in the epidermis of plants. They occur mostly in leaves.

Structure of Stomata:

Each stomata is surrounded by two kidney shaped guard cells containing chloroplast, which the opening & closing of stomata and the inner wall of the guard cell is thick

and inelastic due to presences of a secondary cellulose layer. The outer wall of the guard cell is thin, elastic and permeable.

Opening and Closing of Stomata:

The opening & closing of stomata depends upon the turgidity of their guard cells.

Opening of Stomata:

Stomata are open when the guard cells are turgid. The inner margins curve a part due to outward stretching of outer cells resulting in the acidifying of stomata pore.

Closing of Stomata:

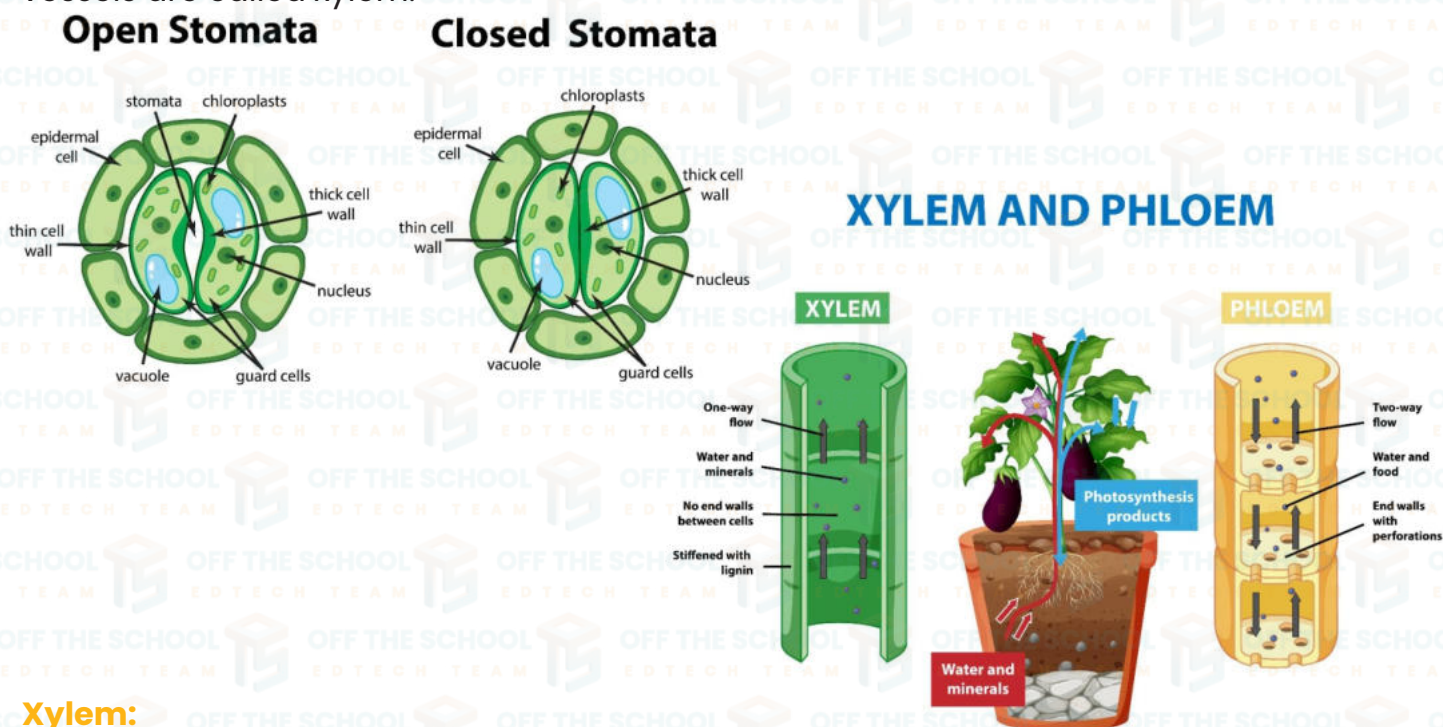
Stomata are closed when the turgidity of guard cells decreases with the loss of turgidity the nine margins become straight and come close to each other which results in the closing of stomatal pore.

Control Mechanism of Opening & Closing of Stomata:

The changes in the turgidity of the guard cells control the opening & closing of stomata. The turgidity of a guard cell is regulated by concentration of solutes present in it which mainly depend upon the rate of photosynthesis. The system involved in the transportation of various substances within the blood of an animal is called circulatory system.

Transport of Water in Stem

Following plants have a system of vessels for transport of water in the stem. These vessels are called xylem.



Xylem:

Xylem tissue consists of vessel elements and tracheid's. Vessel elements are long tubes while tracheid are long cells with overlapping ends.

Xylem Vessel:

A Xylem vessel is a long, hollow, tubular structure from root to leaf; it is made up of many dead cells arranged vertically. The walls of these vessels become strong by the deposition of lignin.

Water and Mineral Transportation:

When a leaf transpires (loses water), the water concentration of its mesophyll cells drops. This drop causes water to move by osmosis from the xylem of leaf into mesophyll cells. When one water molecule moves up in the xylem of the leaf it creates a pulling force that continues all the way to root, this pulling force created by the transpiration of water

is called transpiration pull.

Following are the reasons for the creation of transpiration pull.

- Water is held in a tube (xylem) that has a small diameter.
- Water molecules adhere to the walls of the xylem of xylem tube (adhesion).
- Water molecules cohere to each other (cohesion).

These attractions make an overall tension among water molecules. This tension forms columns of water. The columns of water move from root to shoot and the water content of the soil enters in these columns.

Transportation of Food in Plants:

Phloem:

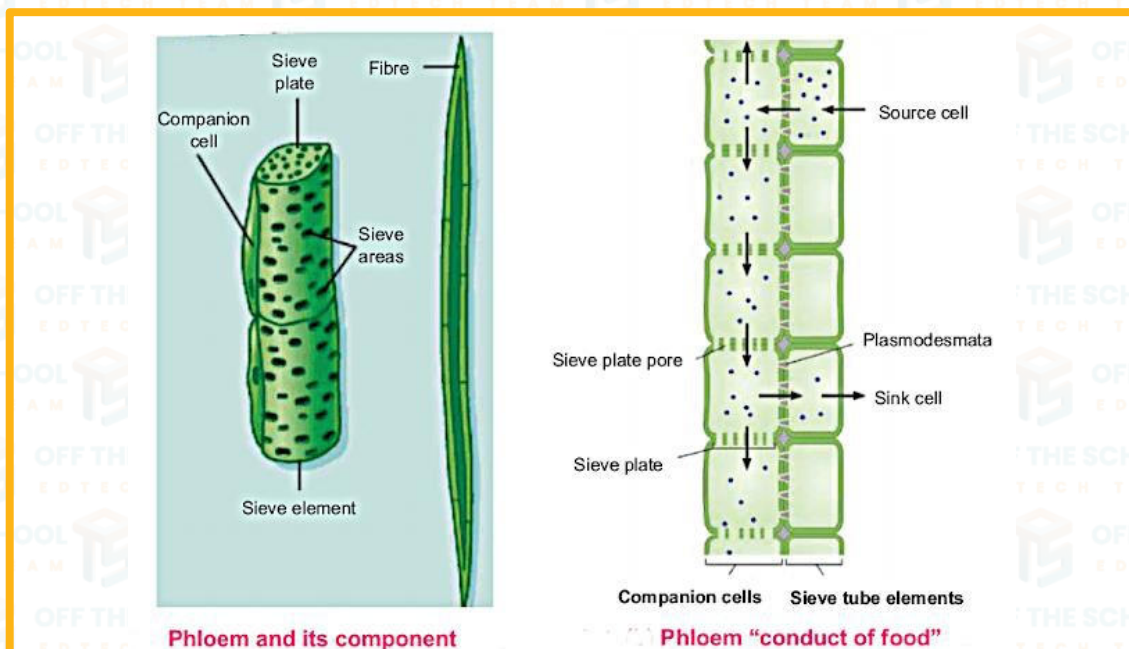
The tissue that transports nutrients like amino acids and sugar from leaves to various parts of plants is called phloem. Phloem is a complex, food conducting tissue in vascular plants. In dicots they are located outside the xylem.

Structure:

- Phloem is composed of four elements:
- Sieve tubes are elongated living cells with perforated end walls, which do not possess nucleus and are main conducting cells.
- Companion cells which are closely associated with sieve tubes and control the sieve tubes.
- Phloem parenchyma which is mainly concerned with storage of organic matter, tannins etc.
- Phloem fibers are elongated cells with lignified walls.

Functions:

They transport food and other metabolic products from leaves to various parts of the plant. This transport is called translocation of food.



CIRCULATORY SYSTEM:

Definition:

The system involved in the transport of various substances within the body of an animal is called Circulatory System. The circulatory system transports gases like O_2 , CO_2 etc. nutrients, wastes, hormones and defense proteins.

Types of Circulatory system:

There are two types of Blood Circulatory Systems found in animals:

- i. Open circulatory system
- ii. Closed circulatory system

(i) Open circulatory system:

In this type of Circulatory System blood flows through the spaces among tissue so it directly comes in contact with tissues. It remains filled in the open tissue spaces called Sinuses. After exchange of materials with tissues, blood enters the pumping organs or heart which pumps it into blood vessels. These vessels again drain out blood into sinuses so it remains in Circulation.

Example:

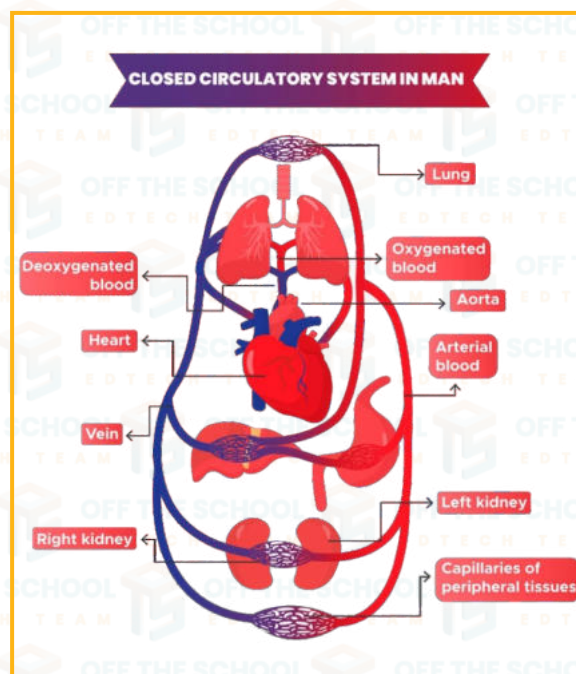
This type of circulatory system found in arthropods and mollusca.

(ii) Closed circulatory system:

This type of system allows blood to flow inside the closed tubular blood vessels and never comes out in direct contact with tissues.

Example:

Human and Most vertebrates and some invertebrates such as, Annelid like earthworm, have a closed circulatory system.



Transport in Man

Circulatory System in Man:

In man, a closed type of circulatory system is found, which consists of the following components.

Blood:

A fluid with cells, substances & other dissolved.

Heart:

A pumping pulsatile organ.

Blood Vessels:

Blood vessels are tubes i.e. Arteries, Veins, and capillaries.

Blood:

Introduction:

It is composed of a liquid called blood plasma & blood cells.

Weight:

Weight of blood in our body is about 1/12th of our body.

Average Blood Volume:

Average adult body has about 5 liters of blood.

Composition of Blood:

In a healthy person, plasma constitutes about 55% and cell or cell like bodies about 45% by volume of the blood.

Chemical Composition of Blood Plasma:

Water constitutes about 90–92% of plasma & 8–10% are dissolved substances.

Erythrocytes or Red Blood Cells (Rbc):

Number:

A cubic millimeter of blood contains 5 to 5.5 million red blood cells in males and 4 to 4.5 million in females.

Loss of Nucleus & Other Organelles:

In the RBCs of mammals, the nucleus is lost when they get mature.

Composition of Cytoplasm of Rbcs:

95% of the cytoplasm of RBCs is filled with hemoglobin which transports O_2 & CO_2 , the remaining 5% consists of enzymes & other proteins.

Shape:

RBCs are biconcave & have an elastic cell membrane.

Formation:

In the embryonic and foetal life they are formed in the liver and spleen. In adults, they are formed in the red bone marrow of short & flat bones, such as the sternum, ribs & vertebrae.

Average Life Span:

The average life span of RBCs is 120 days.

White Blood Cells or Leukocytes (Wbc's):

The average lifespan of red blood cell is about four months (120 days) after which it breaks down in liver & spleen by phagocytosis

White Blood Cells or Leukocytes (Wbc's):**Introduction:**

These are colourless.

Location:

They are not confined to blood vessels only as they also migrate out into the tissue fluid.

Number:

One cubic millimeter of blood contains 7000 to 8000 WBC's

Life Span:

Their life span ranges from months to even years, depending on the body's needs.

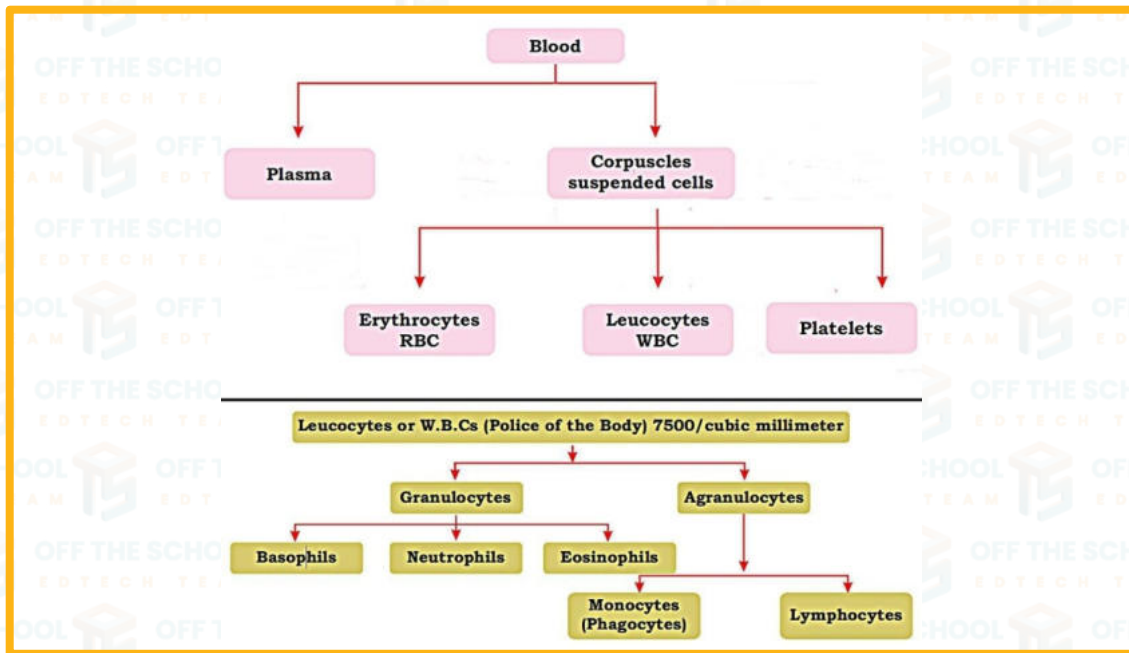
Function:

WBC's function as the main agents in the body's defense system.

Types of Leukocytes:

Leukocytes are divided into two main types.

- i. Granulocytes
- ii. Agranulocytes



Platelets (Thrombocytes):

Introduction:

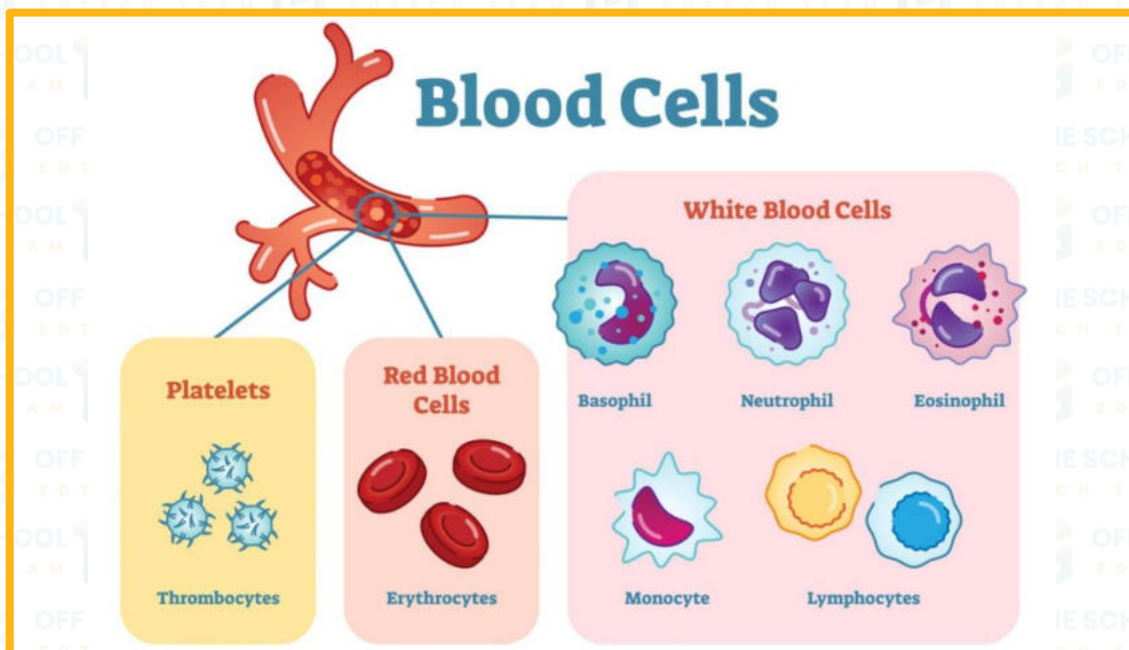
They are not cells but are fragments of large cells of bone marrow called megakaryocytes they do not have any nucleus & any pigment.

Number:

One cubic millimeter of blood contains 250000 platelets.

Function:

They play an important role in blood clotting. The clot serves as a temporary seal at the damaged area.



Function of Blood:

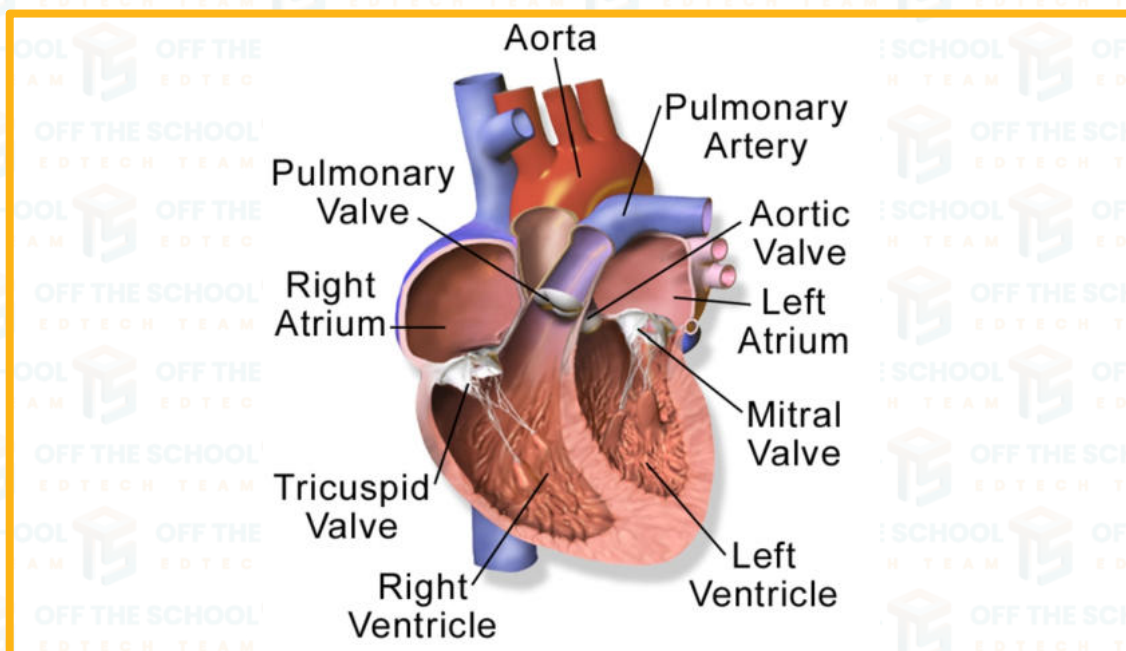
1. It transports the gases to the tissues and from the tissues.
2. It defends the body by killing the germ which somehow enters the body.
3. It transports the nutrients from the gut to all parts of the body.
4. It transports the nitrogenous wastes from tissues to excretory organs.
5. It transports the hormones from the endocrine glands to their target organs.

6. It maintains a uniform body temperature.
7. It makes a clot by itself to stop the bleeding.

Structure of Human Heart

Heart is a muscular, pumping organ, it is located in the thorax. It is enclosed in a fibrous bag like protective cover called pericardium. Externally it is conical in shape. Internally it consists of four chambers.

1. Right atrium
 2. Left atrium
 3. Right ventricle
 4. Left ventricle
- 1) Upper two chambers are thin walled atria while the lower two are thick walled ventricles.
 - 2) Atria are completely separated from each other by an interatrial septum.
 - 3) Ventricles are separated by a muscular partition called inter ventricular septum.
 - 4) An auricular-ventricular aperture connects each atrium with the ventricle of its own side.
 - 5) In between the right atrium and right ventricle lies a tricuspid valve.
 - 6) Similarly, a bicuspid valve guards the opening between the left atrium and left ventricle.
 - 7) These valves prevent the backflow of blood from the ventricles to the atria.
 - 8) Two atria contract to flow the blood into ventricles which simultaneously push the blood into the body.
 - 9) The blood from the right ventricle is pumped through the pulmonary area of the lungs for oxygenation while blood from the left ventricle is pumped through the aorta to all parts of the body.



Double Circuit Circulation:

The circulation of blood in our body is called double circuit circulation because it circulates in two separate circuits.

- 1) Pulmonary circuit
- 2) Systemic circuit

Pulmonary Circuit:

- The right atrium receives deoxygenated blood from all parts of the body (except

lungs) through two large veins called Superior and Inferior Vena Cava respectively.

- This deoxygenated blood passes into the right ventricle through the tricuspid valve.
- The right ventricle upon contraction forces the deoxygenated blood into the pulmonary arch which bifurcates into pulmonary arteries immediately after leaving the heart.
- Each pulmonary artery supplies the blood (deoxygenated) to the lungs of its side.
- While passing through the capillary network of lungs, the blood gives up its carbon dioxide and absorbs oxygen.
- The oxygenated blood comes back to the left atrium through a pair of pulmonary veins from each lung.

This entire circuit i.e. blood from the lungs to heart and heart to lungs is called pulmonary circuit.

Systemic Circuit:

It is responsible to supply oxygenated blood to all parts of body except lungs through systemic arch or aorta. This is called a systemic circuit. Initially, the aorta gives off three branches which supply blood to the head, arms and shoulders. The aorta then descends down and gives off many branches which supply blood to their respective organs e.g.

i. Hepatic artery – supplies blood to liver

ii. Renal artery – supplies blood to kidneys

iii. Gastric or Mesenteric arteries – supply blood to the stomach intestine.

The deoxygenated blood is collected from all organs by the blood vessels called veins e.g.

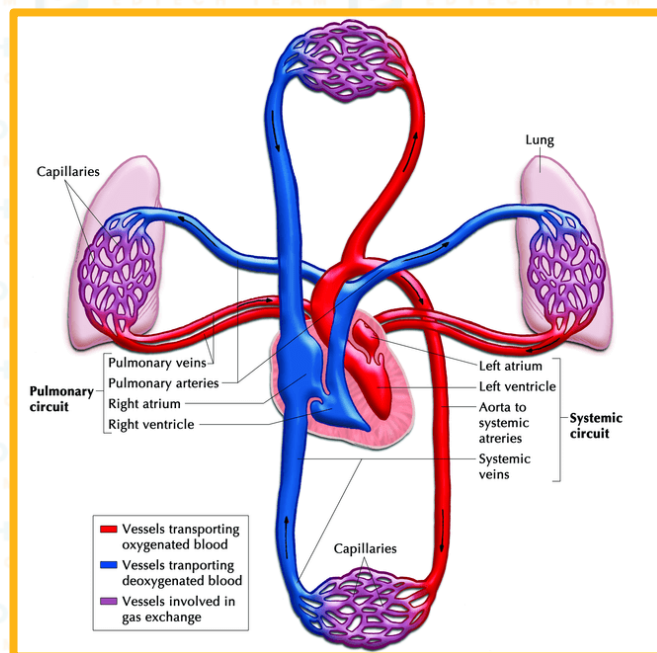
i. Renal veins – collect deoxygenated blood from kidneys

ii. Hepatic portal vein – collects deoxygenated blood from the alimentary canal.

iii. Hepatic vein – collects deoxygenated blood from liver.

All these veins from posterior part of the body fuse together to form posterior vena cava. The veins from all the organs anterior to heart i.e. head and forelimbs fuse to form anterior vena cava.

Both anterior and posterior vena cava discharge all the venous blood into the right atrium from where through right ventricle it is pumped to lungs through pulmonary circuit for oxygenation.



Arteries:

The blood vessels that carry blood from heart to various organs of the body are termed

as arteries. They are thick walled and more elastic than veins. Due to their thick walls, their inner space or lumen is narrower. They lie deep in the body. When an enters in corresponding organ it divides up into smaller branches called arterioles. The arteriole eventually divides into a fine microscope interconnected network of capillaries.

Structure:

The wall of an artery is composed of three layers.

- i. An inner layer of endothelial cells.
- ii. A thick middle layer of smooth muscles and elastic fibers.
- iii. An outer layer of elastic fibers and connective tissue.

Capillaries:

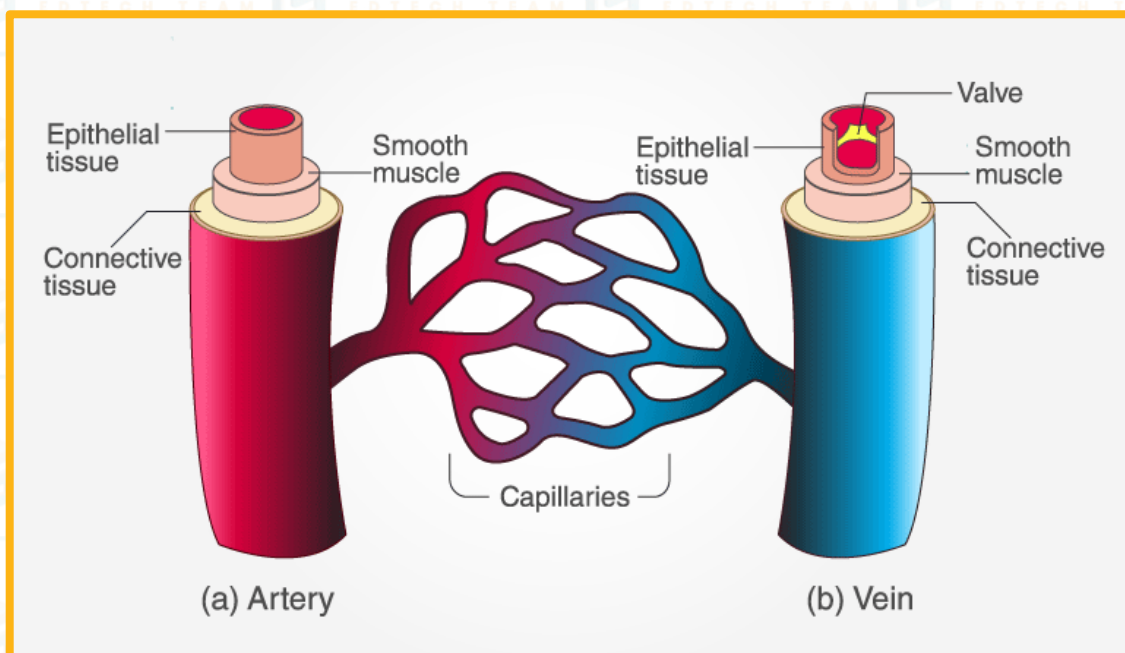
There are microscopic vessels where exchange of various substances occurs between blood and the surrounding tissues. Each capillary is composed of only single cell layers of endothelial cells. Internally, its diameter is slightly larger than that of a R.B.C. They are connected with arterioles on one side and vessels on the other side.

Veins:

Veins are the blood vessels that bring blood back to heart. They are formed by the union of smaller branches called venules which in turn are formed by the fusion of capillaries with each other. The blood flow in a vein is slower and uniform as composed to the artery.

Structure:

Veins are thin walled and less elastic than arteries. The walls of a vein are also composed of the same three but less muscular layers.



Differentiate Between Arteries and Veins:

S.no	ARTERIES	VEINS
01	Arteries are the blood vessels that carry blood from heart to various organs of the body.	Veins are the blood vessels that bring blood back from various organs to the heart.
02	Arteries are thick walled and more elastic.	Veins are thin walled and less elastic.

03	The walls of arteries are more muscular.	The walls of veins are less muscular.
04	They lie deep in the body.	They do not lie deep in the body.
05	Arteries divide up into small branches called arterioles.	Veins divide up into small branches called venules.